

Credit Risk Modeling: From Model to Business Value

Abstract

This project builds a credit risk model using Home Credit data, aiming to achieve both accuracy and stability over time. Using systematic preprocessing, feature engineering, and a LightGBM model optimized for PR-AUC under class imbalance, the model shows strong predictive power (ROC-AUC above 0.85) and stable results across different periods. From a business view, the model can be used with confidence for credit approval and monitoring, keeping portfolio risk under control across different bad-rate caps. Most top features are stable, but two unstable drivers were found, which should be monitored to avoid future model drift.

1. Data

1.1 Source

The dataset used in this project originates from the Kaggle [competition Home Credit – Credit Risk Model Stability](#). It contains historical loan application records from Home Credit, with the target variable indicating whether the client experienced payment difficulties (1 = default, 0 = non-default). All experiments are based solely on the train set provided by the competition in Parquet format.

1.2 Structure

- Observation unit: each credit case. (case_id)
- Target variable: target (binary default indicator)
- Temporal index: WEEK_NUM, used for time-based model evaluation and stability checks
- Feature origins: application-level static features, historical behavioral features, and derived transformations
- Feature suffixes denote transformation types. **P** (DPD (Days past due)), **A** (amounts), **D** (dates), **M** (masked categorical variables) and **T** and **L**(Unspecified Transform).

1.3 Data Characteristics

- **Imbalance:** Defaults form only a small proportion of applications
- **Temporal drift:** Feature distributions and default rates change across time
- **Missing values:** Many features have high missingness, requiring domain-specific imputation
- **Heterogeneity:** Data come from multiple historical tables, mixing numeric, categorical, and date-derived features

2. Data Preprocessing – Stage 1 (Before Merge)

This stage is performed before merge and splitting, when the dataset still contains multiple historical records (rows) per case_id from depth=1 and depth=2 tables. All processing here is row-level within historical records of case_id.

2.1 Missing Value Imputation

Applied suffix-based imputation rules:

- Numeric amount or DPD variables → fill with 0.
- Masked categorical variables → fill with "missing" (string) or -1 (numeric category codes).
- Date and certain mixed-type variables → left unfilled to preserve raw information.

2.2 Additional Feature Creation

Generated domain-driven combination features at the historical record level, such as delinquency ratios, debt-to-income measures, credit utilization, repayment progress, and overdue structure indicators.

Constructed row-level domain-driven combination features within each historical record, including:

- **Delinquency behavior:**
 - Overdue Days to Balance Ratio, Recent Delinquency Worsening Indicator, Severe Delinquency (≥ 60 Days) in Past Year/Two-Year, Chronic Severe Delinquency (≥ 60 Days) over 24 Months, Delinquency Improvement Indicator, Early-Stage Delinquency Indicator
- **Debt & income capacity:**
 - Debt-to-Income Ratio (DTI), Disposable Income Ratio, Total Debt Burden Ratio, High Leverage Indicator (DTI > 60%)
- **Credit usage:**
 - Credit Utilization Ratio, Credit Card Balance-to-Limit Ratio
- **Repayment & overpayment:**
 - Repayment Progress Ratio, Overpayment Indicator
- **Overdue structure:**
 - Active Account Overdue Ratio, Closed Account Overdue Ratio, Active Account Overdue Indicator, Closed Account Overdue Indicator, Payment-Level Overdue Indicator
- **Other financial behavior:**
 - Payment Ability Ratio, Residual Balance-to-Price Ratio (Active Accounts), Residual Balance Indicator (Closed Accounts), Revolving Account Ownership Indicator, Delinquency Tolerance Ratio, Recent-to-Historical Delinquency Relief Ratio, Short-Term vs. Long-Term Delinquency Delta, Severe Current Delinquency Indicator(≥ 60 Days), Negative Delinquency Change Indicator

These features capture payment history, credit utilization, repayment patterns, and overdue structures at the historical record level.

2.3 Aggregation of Historical Data

For depth=1 and depth=2 tables, aggregated historical records per case_id according to suffix/type-specific rules:

- **Numeric amount / DPD features (P, A):** max, last, mean.
- **Date features (D):** converted to days from date_decision, then max, last, mean.
- **Categorical (masked) features (M):** max, last.
- **Other transformed features (T, L):** max, last.
- **Index count features (num_group columns):** max, last.
- **Flag / Ratio / Delta features (combo feature):** mean, max, last.

This aggregation logic compresses multiple historical rows per case_id into one case-level row while preserving both recent and long-term behavioral signals.

2.4 Merge & Conflict Resolution

Sequentially merged aggregated tables into the base table on case_id. For conflicting columns, values from the left table were prioritized, with nulls filled from the right table.

2.5 Date Features

Extracted decision month and weekday, and converted all _D date variables into days relative to date_decision.

3. Data Split

To keep the evaluation realistic, the dataset was split by decision week. After cleaning data types and sorting records by WEEK_NUM and weekday_decision, I built three non-overlapping subsets:

- **Training set (Weeks 0–69):** used to train the model and perform feature selection.
- **Validation set (Weeks 70–74):** used to search for the business threshold.
- **Test set (Weeks 75–91):** used to evaluate both model and business performance.

This chronological split makes sure the model is tested in the same way it would face real future cases.

4. Data Preprocessing – Stage 2 (After Split)

This stage is performed after aggregation and chronological splitting, when the dataset is already in one row per case_id format. All processing here is application-level with aggregated case-level features.

4.1 Missing Value Imputation

Applied a second round of suffix-based imputation to handle any remaining missing values in the aggregated case-level dataset, after applying a hard filter (removing features with missing value ratio > 97% or cardinality > 5000).

4.2 Additional Post-Split Features

Constructed post-split case-level combination features using only data available up to each dataset's time period, including:

- High Utilization with Severe Delinquency Indicator – flag for high credit card utilization and severe delinquency co-occurring.
- Recent Delinquency Change – recent actual DPD minus its historical mean.
- Credit Card Burden Relief Ratio – relative improvement in credit card burden.
- Excess Debt-to-Income Ratio – annuity relative to main occupation income.
- Arrears Progression Indicator – interaction of repayment progress and delinquency severity.
- Residual Balance Indicator – combined residual amount and residual closed account indicators.
- Credit Limit Change – change in credit limit from historical mean.
- No Recent Installment Payment Indicator
- Zero Debt-to-Income Indicator

These features are designed to capture recent shifts and interactions in credit behavior that may not be visible from pre-split row-level combos.

4.3 Winsorization of Extreme Values

To reduce the influence of outliers, numeric variables were winsorized at the 99.8th percentile, based on the training set distribution. The same upper limits were applied to the validation and test sets to keep the datasets comparable to prevent any information leakage. In the credit risk setting, winsorization reduces the effect of extreme amounts or credit exposures, while keeping the relative order of cases' risk levels.

4.4 Encoding of Categorical Variables

Categorical predictors were processed with two approaches:

- **One-hot encoding** was applied to low-cardinality features (with ≤ 20 distinct categories).

- **Target encoding with temporal ordering** was applied to high-cardinality variables (with *> 20 distinct categories*). For each record, the encoding was calculated only from past data available up to that point, using a rolling time window, which keeps the time-dependent nature of credit risk, prevents forward-looking bias, and helps the model reflect changing default risks across borrower groups.

4.5 Correlation Pruning

Removed highly correlated features (higher than 0.97) based on train set only using a preference hierarchy (mean = 1, max = 2, last = 3) that retained more stable and interpretable aggregations.

4.6 Feature Selection (Gain + SHAP Union)

Since the modeling approach is based on **GBDT**, I did not use traditional credit risk methods such as WOE and IV, which are mainly designed to capture linear relationships for logistic regression. Instead, I applied **gain-based feature importance** and **SHAP** for feature selection. I performed supervised feature selection on the training set only. A gradient-boosted tree model (LightGBM) was fitted to obtain two important views: (i) gain-based feature importance and (ii) mean absolute SHAP values. I then retained the smallest feature sets that cumulatively explain 95% of total gain and 90% of total mean absolute SHAP values, respectively, and took their union. The resulting union was applied unchanged to the validation and test sets, preventing temporal leakage while balancing model-driven relevance and attribution robustness.

5. Model Development

I applied a **time-series expanding window cross-validation**, where each validation set followed the chronological order of loan origination. The validation windows had a fixed length and were arranged sequentially near the end of the training period. For example, with an end point at week 69 and a validation span of 4 weeks in this project, the validation windows were [54–57], [58–61], [62–65], and [66–69]. In each case, the training window always covered all weeks from the start up to the week before the validation period, which ensured an expanding training set.

The model was optimized for **PR-AUC**, highlighting the importance of ranking performance under severe class imbalance. Hyperparameters were tuned automatically with **Optuna**, and class imbalance was handled by **re-weighting positive samples**.

The final model was then trained on the full training period (weeks 0–69) with the selected parameters and stored for later stability checks and business evaluation.

6. Model Evaluation

6.1 Model Performance

I evaluated the model’s predictive performance using multiple metrics, including ROC-AUC, PR-AUC, and KS statistics.

- **Overall Performance:** The aggregated metrics are summarized in Table 1 (Overall Model *Information*), which shows consistently high ROC-AUC values across validation and test sets, reliable KS statistics, and stable PR-AUC despite class imbalance.
- **Temporal Stability:** Weekly performance plots (Figure 1 Weekly Model Performance – Validation) and (Figure 2 Weekly Model Performance – Test) — show that all three main metrics remained relatively stable. Importantly, no downward trend was observed, suggesting that the model remains robust when applied to future data.

split	the number of case	Positive rate	pr_auc	roc_auc	ks	brier	ece
TRAIN	1302727	0.0331	0.2593	0.9013	0.6495	0.1429	0.2682
VAL	40554	0.0236	0.1756	0.8569	0.5550	0.1165	0.2380
TEST	183378	0.02158	0.1845	0.8718	0.5855	0.1488	0.2869

Table 1. Overall Model Information

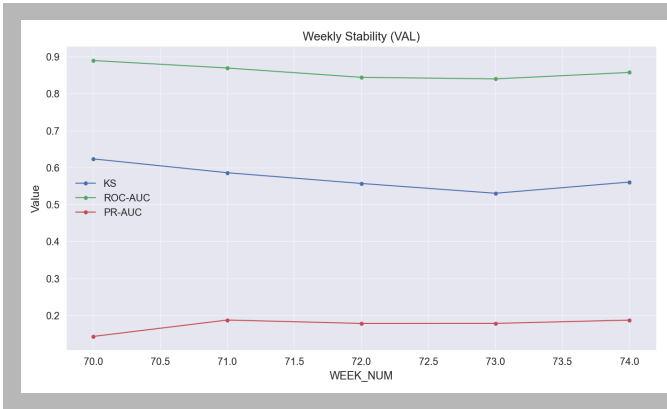


Figure 1 Weekly Model Performance – Validation

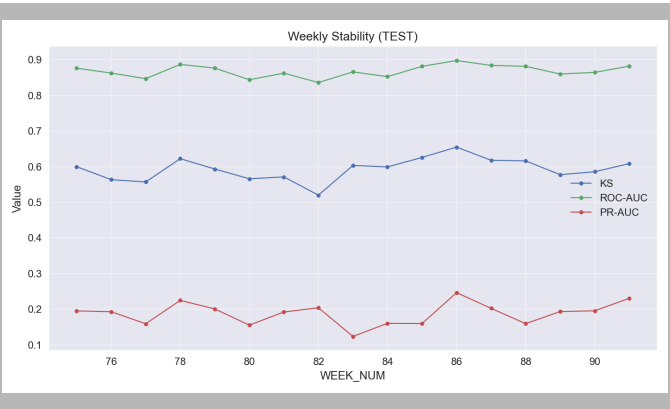


Figure 2 Weekly Model Performance – Test

From a business perspective, the results suggest that the model can be deployed with confidence for credit approval and monitoring. Its stable performance over time reduces concerns about sudden drops in accuracy, helping to maintain portfolio quality and support long-term risk management goals. If the model’s KS declines in the later part of the test period, this may indicate potential overfitting and should be investigated further.

6.2 Stability Analysis

To assess the temporal robustness of the model, I computed **Population Stability Index (PSI)** by comparing score distributions between training, validation, and test datasets. Results are presented in two layers: overall stability and weekly stability.

6.2.1 Overall PSI

- **Validation (Weeks 70–74 vs TRAIN):** Average PSI = 0.0567
- **Test (Weeks 75–91 vs TRAIN):** Average PSI = 0.0373

All values are well below 0.1, indicating stable score distributions without evidence of systematic drift.

6.2.2 Weekly PSI

- **Validation Period:** Weekly PSI ranges from 0.047 to 0.067, consistently stable across Weeks 70–74. (Figure 3 *Weekly PSI(VAL to TRAIN)*)
- **Test Period:** Weekly PSI fluctuates between 0.03 and 0.101, with a temporary spike near Week 79 but no persistent drift. (Figure 4 *Weekly PSI(TEST to TRAIN)*).

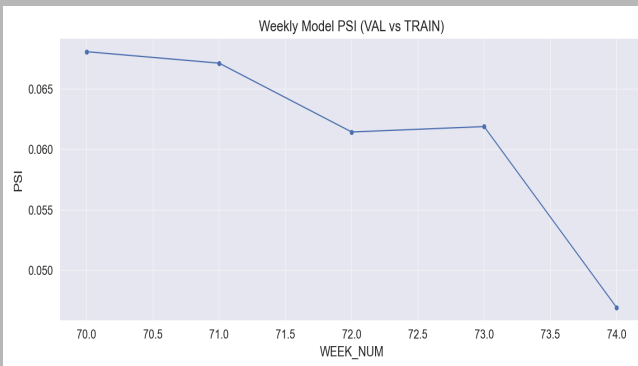


Figure 3 Weekly PSI(VAL to TRAIN)

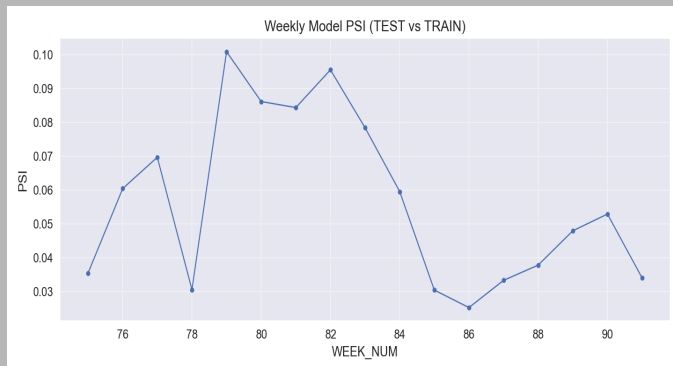


Figure 4 Weekly PSI(TEST to TRAIN)

Overall, weekly stability analysis confirms that the model maintains acceptable consistency, with only short-term variations observed.

7. Business Evaluation

7.1 Threshold & Approval Analysis

To translate model scores into actionable credit decisions, I analyzed the relationship between threshold, approval rate, and bad rate.

- On the **validation set**, the threshold–approval–bad-rate curve (Figure 5 Threshold-VAL) was used to identify a cut-off that balances portfolio growth with risk control.
- On the **test set**, the same cut-off was applied, and its performance was validated (Figure 6 Threshold-TEST).

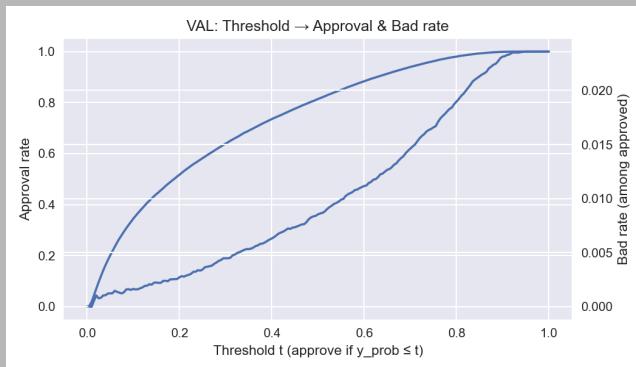


Figure 5 Threshold-VAL



Figure 6 Threshold-TEST

This approach ensures that threshold selection is not overly tuned to historical data and remains effective when generalized to unseen periods.

7.2 Discriminatory Power (Lift & Gain)

To evaluate business value, I assessed the model's ability to prioritize high-risk clients.

- The **Lift Curve** (Figure 7 Lift Curve) demonstrates that the model achieves significantly higher default capture rates in the top deciles compared to random selection, confirming its discriminatory power.
- The **Cumulative Gain Chart** (Figure 8 Cumulative Gain) further shows that a relatively small fraction of the population accounts for a disproportionately large share of defaults, enabling more efficient allocation of risk exposure.

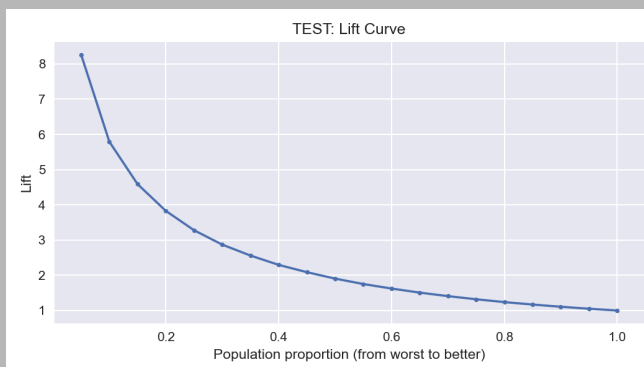


Figure 7 Lift Curve

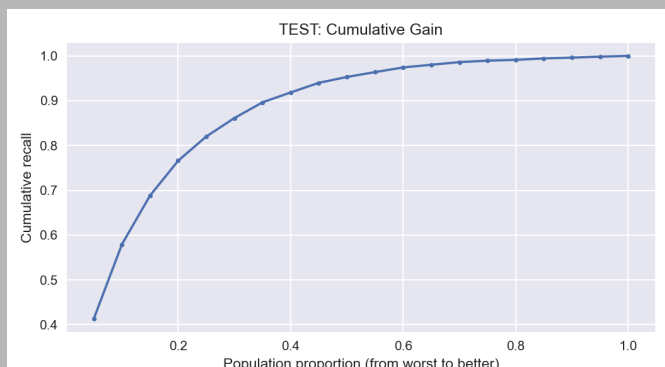


Figure 8 Cumulative Gain

To complement the lift curve, we evaluated the model's performance at the top 5%, 10%, and 15% of predicted risk scores. Table 2 (test_PR_at_K) indicates that the model is highly effective at identifying risky customers in the top-ranked groups.

K%	topN	precision_at_K	recall_at_K
5%	9168	0.1783	0.4132
10%	18337	0.1249	0.5790
15%	27506	0.0990	0.6881

Table 2. test_PR_at_K

These results highlight that the model can materially improve credit decisioning by concentrating risk detection in the early scored segments.

7.3 Strategy Performance (Threshold & Bad-Rate Caps)

To turn the model results into practical business rules, I tested threshold-based approval strategies under different bad rate limits. The **threshold–approval–bad-rate curve** (Figure 5 Threshold-VAL) shows the balance between approval rate and bad rate across score cutoffs on the validation set. This curve helped determine operating points by matching risk tolerance with business needs.

On the test set, the same simulation (Figure 6 Threshold-TEST) shows that the chosen thresholds work well in practice, keeping bad rate under control while still allowing enough approvals.

A summary of the main scenarios is given in Table 3 (Overall business information). Each row represents a candidate threshold, with the related approval rate, bad rate, and default capture rate. The table makes the trade-offs clear: stricter thresholds lower portfolio risk but reduce approval coverage, while looser thresholds expand coverage but increase credit losses.

These findings provide business stakeholders with a clear and practical framework for setting decision rules in line with portfolio-level risk appetite. In addition, the weekly stability analysis attached to the table shows that the strategy's performance stays consistent over time, reducing concerns about temporal drift.

bad_cap	threshold	val_approve_rate	val_bad_rate_in_approved	val_bad_capture_rate	test_approve_rate	test_bad_rate_in_approved	test_bad_capture_rate
0.016	0.7318311	0.9542	0.0160	0.3532	0.9267	0.0116	0.5021
0.018	0.778016	0.9734	0.0180	0.2581	0.9547	0.0138	0.3907
0.02	0.8200516	0.9857	0.0200	0.1651	0.9756	0.0162	0.2661

Table 3. Overall business information

7.4 Stability Analysis

Across the three business caps (1.6%, 1.8%, 2.0%), the weekly bad rate in validation and test sets stayed under control, as shown in Table 4 (Strategy Table with Weekly Stability under Bad-rate Caps).

- At 1.6% cap, stability was strongest: validation went over the cap in 40% of weeks, but the test set never crossed it. The worst single-week bad rate was about 1.58%, with very low variation (std $\approx$ 0.20%).
- At 1.8% and 2.0% caps, stability became looser. The test set breached the cap in 5.9% and 17.6% of weeks. Even then, the highest bad rate ($\sim$ 1.8–2.1%) were only slightly higher than the cap.

bad_cap	threshold	val_weeks_over_cap	val_worst_week	val_std	test_weeks_over_cap	test_worst_week	test_std
0.016	0.7318	0.4	0.0184	0.0017	0	0.0158	0.0021
0.018	0.7780	0.2	0.0207	0.0017	0.0588	0.0181	0.0023
0.02	0.8201	0.2	0.0231	0.0018	0.1765	0.0214	0.0027

Table 4. Strategy Table with Weekly Stability under Bad-rate Caps

The stability of the approved population was also assessed using PSI, as summarized in Table 5 (Approved PSI Stability Table). Median weekly PSI stayed below 0.06 across all caps, which is within the “no significant shift” level (<0.1). The maximum weekly PSI values were also below 0.10 (0.087–0.093), showing that even in the worst weeks, the distribution did not change much compared with validation.

Overall PSI values ($\sim$ 0.028–0.035) confirm that the approved population kept stable patterns over time.

bad_cap	threshold	overall_PSI_approved	median_week_PSI_approved	max_week_PSI_approved
0.016	0.7318	0.0277	0.0480	0.0873
0.018	0.7780	0.0310	0.0518	0.0895
0.02	0.8201	0.0346	0.0544	0.0928

Table 5. Approved PSI Stability Table

The weekly stability analysis attached to the table shows that the strategy’s performance stays consistent over time, reducing concerns about temporal drift.

Based on the current analysis, I recommend using a 1.8% cap, as it delivers the highest return under the assumption of equal loan amounts per case. This recommendation holds as long as the per-case profit remains within a normal range. If detailed loan amount data were available, a more refined profit and risk analysis could be carried out.

8. Feature Analysis

8.1 Overall Feature Analysis

First, I used SHAP to identify the ten most important features and carried out an overall analysis. In most cases, the top ten already explain the majority of the model. The combined stability and predictive diagnostics are summarized in Table 6 (Top10 SHAP Feature PSI Stability Table).

On stability (PSI), most features such as **mean_dateofcredstart_739D**, **avgdpdtolclosure24_3658938P**, and **mobilephncnt_593L** remain highly stable across validation and test. In contrast, **mean_refreshdate_3813885D** and **max_amount_4527230A** show extremely high PSI, suggesting severe distribution drift.

On discriminatory power (KS), features like **mean_refreshdate_3813885D** reach very high KS (>0.95), but this strength may mislead based on their instability. More reliable drivers such as **mean_dateofcredstart_739D** and **pmtnum_254L** maintain moderate KS while also being stable.

On information value (IV), most features retain positive and consistent contributions across samples, though **max_sex_738L_M** shows inconsistent IV between validation and test, hinting at sample imbalance rather than genuine signal.

Taken together, the majority of top drivers combine low PSI with steady KS and IV, confirming both stability and predictive reliability. A small number of features, however, are unstable or inconsistent, especially, two features, **mean_refreshdate_3813885D** and **max_amount_4527230A**, which show clear distribution shifts with very high PSI, making them risk signals that require close monitoring or replacement.

feature	psi_train_to_val	psi_train_to_test	ks(train-val)	ks(train-test)	iv(val-train)	iv(test-train)
mean_dateofcredstart_739D	0.0332	0.0448	0.0443	0.0730	0.0610	0.0705
max_sex_738L_M	5.0558	0.0031	0.3795	0.0269	-0.0390	0.0233
mean_refreshdate_3813885D	11.1747	11.1946	0.9577	0.9589	-0.0192	-0.0128
avgdpdtolclosure24_3658938P	0.0018	0.0020	0.0188	0.0274	0.0534	0.0954
mobilephncnt_593L	0.0234	0.0229	0.0679	0.0675	-0.0189	0.0076
mean_dpdmx_139P	0.0029	0.0040	0.0150	0.0306	-0.0141	-0.0140
pmtnum_254L	0.0625	0.0305	0.0879	0.0605	0.0314	0.0468
mean_overdueamountmax_155A	0.0005	0.0011	0.0010	0.0261	0.0199	-0.0282
mean_birth_259D	0.1013	0.1023	0.1360	0.1301	0.1883	0.1107
max_amount_4527230A	1.0493	1.1101	0.3360	0.3381	-0.0054	-0.0163

Table 6. Top10 SHAP Feature PSI Stability Table

8.2 Unstable Feature Analysis

Within the SHAP top 10, we focused on `mean_refreshdate_3813885D` and `max_amount_4527230A` for deeper investigation, as their instability indicates potential model drift and higher business risk if left unmonitored.

8.2.1 `mean_refreshdate_3813885D`

The feature `mean_refreshdate_3813885D` represents the average date when the credit bureau's public sources were last updated, which reflects the recency of external credit data availability. According to Figure 7 (mean SHAP per bin of `mean_refreshdate_3813885D`), the longer the relative time since the last refresh, the lower the risk contribution. This aligns with business logic: if a long period has passed since the last update, it suggests no new overdue events have been recorded, indicating lower risk. In addition, Figure 8 (mean weekly SHAP of `mean_refreshdate_3813885D`) shows that its contribution in validation and test sets remains stable, proving that the feature continues to be effective. The abnormal PSI, KS, and IV values are therefore more likely caused by differences in the data distribution, possibly due to irregular updates in the credit bureau's public sources, rather than by a real loss of predictive power. Thus, this feature should not be discarded but closely monitored in future use.

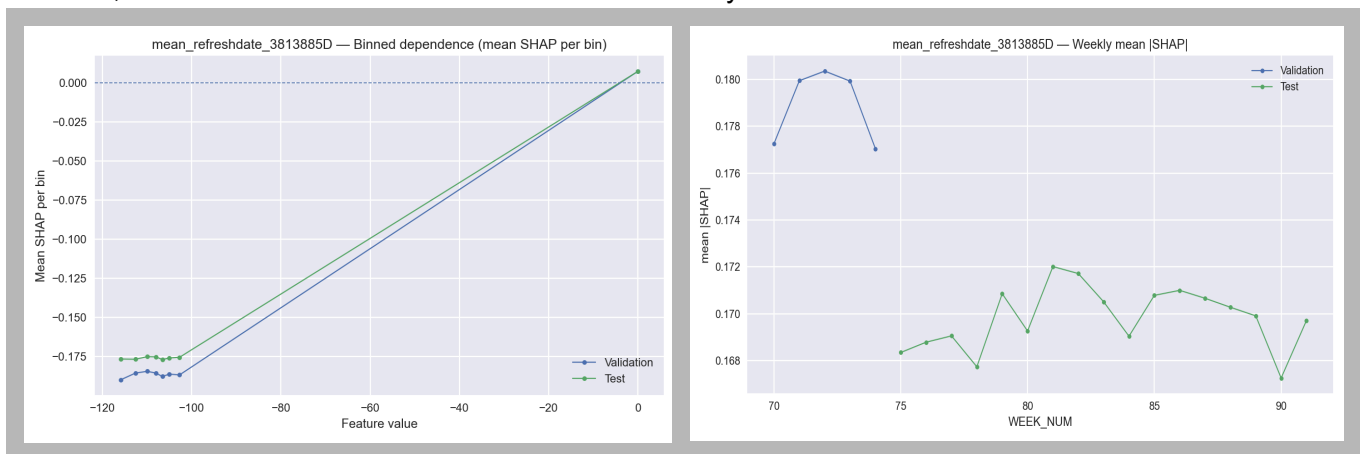


Figure 9 mean SHAP per bin(`mean_refreshdate_3813885D`)

Figure 10 mean SHAP weekly(`mean_refreshdate_3813885D`)

8.2.2 `max_amount_4527230A`

`max_amount_4527230A` represents the largest tax deduction amount tracked by the government registry. According to Figure 9 (mean SHAP per bin), the feature contributes consistently to risk prediction, but the lack of dispersion in values limits its discriminative power. In practice, the largest deduction is often dominated by housing-related items, which explains why most values are concentrated in a narrow range. Figure 10 (weekly mean SHAP) further shows that, although there are fluctuations across test weeks, the contribution remains within a relatively stable band. The high PSI and KS indicate dataset-level distributional shifts, likely reflecting structural changes in how the government registry reports or records deduction amounts, or social factors such as housing price changes, rather than a breakdown of the feature's effect. Thus, this variable should be treated as a risk signal that requires monitoring for reporting consistency and market sensitivity, but not immediately discarded.

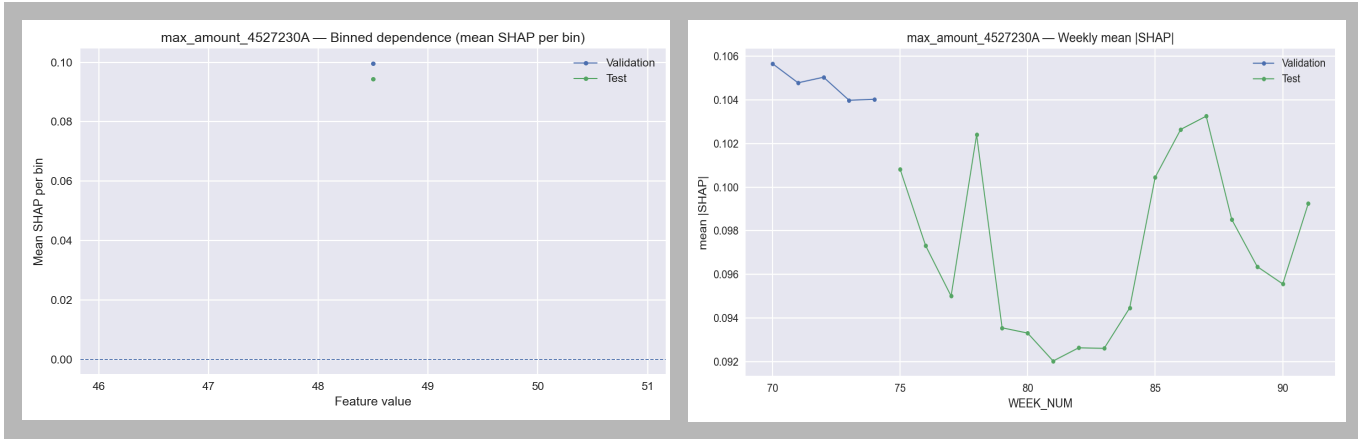


Figure 11 mean SHAP per bin(max_amount_4527230A)

Figure 12 mean SHAP weekly(max_amount_4527230A)

9. Probability Calibration & Uncertainty Control

9.1 Motivation

Although the LightGBM model demonstrates strong ranking performance (ROC-AUC, PR-AUC, KS) and temporal stability (PSI), credit decisioning ultimately depends on **accurate probability estimation**, not only correct ranking.

In practice, approval thresholds, bad-rate caps, capital allocation, and pricing decisions all assume that predicted probabilities can be interpreted as expected default rates. Therefore, probability calibration is essential for reliable deployment.

To address this, I applied **Platt scaling** on the validation set and compared raw vs calibrated probabilities on both validation and test sets.

9.2 Calibration Performance (Raw vs Platt)

The comparison is summarized below:

tag	Brier	ECE
Val(raw)	0.1165	0.2380
Val(cal)	0.0211	0.0019
Test(raw)	0.1488	0.286946647
Test(cal)	0.0194	0.0104

Table 7. Raw Probability VS Calibrated Probability

The improvement is substantial:

- Brier score drops by nearly **85–90%**
- ECE drops by over **95%**
- Out-of-sample improvement (test set) remains consistent

This indicates that while the original GBDT model provides strong ranking performance, its raw probability outputs are significantly miscalibrated (over-confident). After Platt scaling, predicted probabilities align closely with empirical default frequencies.

Importantly, calibration does not alter ranking performance (ROC-AUC / PR-AUC remain unchanged), meaning discrimination ability is preserved while probability reliability is significantly improved.

9.3 Business Implications of Calibration

From a credit risk perspective, calibration directly impacts:

1. **Bad-rate cap control**
Well-calibrated probabilities allow approval thresholds to correspond more accurately to expected portfolio bad rates.
2. **Capital allocation**
If predicted PD can be interpreted as expected default rate, provisioning and capital planning become more reliable.
3. **Pricing & risk-based segmentation**
Miscalibrated probabilities may distort risk-based pricing, especially in low-default portfolios.

The calibrated model therefore enhances deployability and reduces hidden portfolio risk arising from probability misalignment.

While calibration improves the reliability of predicted probabilities, it does not provide formal uncertainty guarantees for individual predictions.

To address this limitation, we further investigate conformal prediction methods for uncertainty quantification under distribution shift.

The detailed methodology and empirical analysis are provided in **Appendix B**.

Appendix B

Conformal Prediction under Distribution Shift

Appendix B presents a methodological extension for distribution-shift-aware conformal inference. The analysis is conducted from a statistical validity perspective rather than an operational deployment perspective.

B.1 Conformal Prediction for Finite-Sample Coverage Guarantees

While probability calibration ensures that predicted default probabilities are numerically reliable, it does not provide formal guarantees on predictive uncertainty. In credit risk decisioning, beyond accurate point estimates, institutions require statistical control over prediction errors at a predefined confidence level. In particular, it is desirable to ensure that prediction errors occur no more than an acceptable proportion α of the time.

Conformal prediction offers a distribution-free framework for uncertainty quantification. Rather than relying on parametric assumptions about the data-generating process, conformal methods construct prediction sets that satisfy finite-sample coverage guarantees. Specifically, under the assumption of exchangeability, conformal prediction ensures that

$$P(Y \in \Gamma(X)) \geq 1 - \alpha,$$

where $\Gamma(X)$ denotes the prediction set and α is a user-specified error level. The exchangeability assumption requires that the joint distribution of calibration and test observations is invariant under permutations. Formally, for observations $Z_1, \dots, Z_n$, exchangeability implies

$$(Z_1, \dots, Z_n) \stackrel{d}{=} (Z_{\pi(1)}, \dots, Z_{\pi(n)})$$

for any permutation π . This condition implies that calibration and test samples are drawn from the same underlying data-generating distribution and that their ordering does not contain additional information.

Under exchangeability, conformal prediction provides exact finite-sample validity without imposing distributional assumptions on the predictive model. In the following sections, we construct a standard Mondrian conformal baseline using calibrated probabilities and empirically evaluate its coverage behavior under a week-based temporal split.

B.2 Baseline Conformal: Raw vs Calibrated Probabilities

To isolate the impact of probability calibration on uncertainty quantification, we first construct a standard split conformal classifier without any class partitioning or shift-aware adjustments.

B.2.1 Nonconformity Score

For binary classification, we adopt the standard symmetric nonconformity score:

$$s(x_i, y_i) = \begin{cases} \hat{p}(x_i), & \text{if } y_i = 0 \\ 1 - \hat{p}(x_i), & \text{if } y_i = 1 \end{cases}$$

This score measures how incompatible the predicted probability is with the true label. Larger values indicate poorer conformity.

The global threshold is computed as:

$$q = \text{Quantile}_{1-\alpha} \{s(x_i, y_i)\}_{i \in \mathcal{D}_{cal}}$$

where $\mathcal{D}_{cal}$ denotes the calibration set.

For a new observation x , the prediction set is defined as:

$$\Gamma(x) = \{y \in \{0,1\} : s(x,y) \leq q\}$$

Under exchangeability, this construction guarantees marginal coverage at level $1-\alpha$.

B.2.2 Raw vs Calibrated Probability Comparison

We apply the above conformal procedure using:

1. Raw LightGBM predicted probabilities
2. Platt-calibrated probabilities

Data type	Alpha	Coverage	Singleton rate	Avg_set_size
Raw	0.5	0.9214	0.6415	1.3585
Calibrated	0.5	0.9320	0.9442	0.9442

Table 8. Conformal Performance (Raw Probability VS Calibrated Probability)

The results show that probability calibration improves conformal performance:

- Test coverage increases from 0.9214 to 0.9320.
- Singleton rate increases substantially, indicating improved decisiveness.

However, empirical coverage remains below the nominal 95% level.

This suggests that while calibration improves probability reliability and enhances conformal efficiency, it is insufficient to fully restore nominal coverage in the temporal evaluation setting. In the subsequent sections, all conformal analyses are conducted using calibrated probabilities to isolate the impact of distributional discrepancies on coverage performance.

B.3 Baseline Comparison: Unconditional vs. Class-Conditional Conformal

B.3.1 Methodological Extension: Class-Conditional (Mondrian) Conformal

The unconditional conformal classifier treats all calibration samples as exchangeable without stratification. While this guarantees marginal coverage under exchangeability, it does not explicitly control coverage across different outcome classes.

In imbalanced credit portfolios, such marginal treatment may lead to heterogeneous coverage across defaulters and non-defaulters.

To address this, we consider a class-conditional (Mondrian) conformal construction, where nonconformity thresholds are computed separately for each label:

$$q_0 = \text{Quantile}_{1-\alpha} \{s(x_i, y_i) : y_i = 0\},$$

$$q_1 = \text{Quantile}_{1-\alpha} \{s(x_i, y_i) : y_i = 1\}.$$

The prediction set is then defined as:

$$\Gamma(x) = \{y \in \{0, 1\} : s(x, y) \leq q_y\}.$$

Under exchangeability, this construction provides class-conditional coverage guarantees, ensuring that each class achieves coverage at level $1-\alpha$.

In our implementation, all thresholds are estimated under a rolling calibration scheme, where the preceding W weeks are used to compute the quantiles for each test week.

Table 9 reports the rolling evaluation results for two baseline conformal constructions using calibrated probabilities:

Method	coverage	coverage($y = 0$)	Coverage($y = 1$)
Mondrian	0.9430	0.9427	0.9555
Unconditional	0.9455	0.9664	0

Table 9. Conformal Performance (Mondrian VS Uncondition)

Several important observations emerge.

First, both methods exhibit systematic under-coverage relative to the nominal 95% level. The unconditional construction achieves an overall coverage of 0.9455, while the class-conditional Mondrian achieves 0.9430. Both remain below the target level of 0.95.

Second, a critical difference arises when examining class-conditional coverage. While the unconditional conformal classifier achieves high coverage for non-defaulters (Coverage($y = 0$) = 0.9664), it completely fails for defaulters (Coverage($y = 1$) = 0). This indicates that the unconditional construction collapses minority-class coverage under severe class imbalance, rendering it unsuitable for credit risk settings where accurate protection of the default class is essential.

In contrast, the Mondrian construction restores class-conditional validity for the default class, achieving Coverage($y = 1$) = 0.9555 while maintaining Coverage($y = 0$) = 0.9427. This

demonstrates that label-stratified thresholding is necessary to prevent systematic failure on the minority class.

However, despite correcting the class-conditional imbalance, the Mondrian construction does not restore nominal overall coverage. The remaining coverage gap suggests that the primary source of deviation is not label imbalance alone. Instead, other forms of distributional discrepancy across time may be driving the violation of exchangeability.

In the next section, we formally investigate potential sources of distribution shift across the temporal evaluation period.

B.4 Diagnosing Distribution Shift in Temporal Credit Data

The systematic under-coverage observed in Section B.3 indicates a violation of the exchangeability assumption required for finite-sample validity of conformal prediction. Since exchangeability implicitly assumes that calibration and test samples are drawn from the same data-generating process, temporal evaluation in credit portfolios naturally challenges this assumption.

For clarity of analysis, we decompose the potential sources of coverage instability into three components:

1. Temporal covariate shift
2. Temporal conditional shift (concept shift)
3. Cross-sectional conditional heterogeneity

Each mechanism affects conformal validity differently and motivates a different mitigation strategy.

B.4.1 Temporal Covariate Shift

Temporal covariate shift refers to changes in the marginal feature distribution over time:

$$P_t(X) \neq P_s(X), \text{ while } P(Y|X) \text{ remains stable}$$

In credit portfolios, such shift may arise from:

- Changes in borrower demographics
- Policy adjustments affecting applicant composition
- Macroeconomic conditions influencing credit demand

Even if the conditional risk function $P(Y|X)$ remains stable, conformal validity can deteriorate because conformity score quantiles are estimated under $P_s(X)$ but evaluated under $P_t(X)$. If high-risk or low-risk regions are over or under represented in calibration relative to test weeks, empirical coverage may deviate from nominal levels.

Mitigation Strategy:

A common mitigation strategy for covariate shift is density-ratio reweighting. In our framework, density ratios are estimated using a domain classifier and incorporated when computing calibration quantiles.

It is important to note that density-ratio reweighting corrects for marginal feature shift but does not address changes in the risk function itself.

B.4.2 Temporal Conditional Shift (Concept Shift)

Temporal concept shift occurs when the conditional distribution between features and targets changes over time:

$$P_t(Y|X) \neq P_s(Y|X)$$

In credit risk modeling, this may arise when:

- Economic downturns increase default rates for the same risk profile
- Structural regime changes alter borrower repayment behavior

Under concept shift, conformity score distributions change across time even if feature distributions remain stable. Since conformal thresholds are derived from historical residuals, they may become misaligned with future score distributions, leading to persistent under or over coverage.

Mitigation Strategy:

A practical mitigation strategy is to adopt a rolling calibration window, whereby conformity scores are computed using only the most recent W weeks. This strategy allows quantile thresholds to adapt dynamically to evolving residual distributions.

However, it is crucial to emphasize:

- Rolling calibration mitigates the impact of slow-moving concept drift.
- It does not eliminate concept shift.

- Full correction would require model retraining or structural model updating.

Thus, rolling conformal prediction should be viewed as an adaptive stabilization mechanism rather than a definitive solution to concept drift.

B.4.3 Cross-Sectional Conditional Heterogeneity (Not a Shift)

The third mechanism is not a temporal shift but a structural property of the data that the distribution of conformity scores varies across regions of X .

Equivalently, conditional variance or residual structure differs across risk levels:

Score distribution $| X = x$ depends on x

For example:

- High-risk borrowers may exhibit larger conformity scores.
- Low-risk regions may display tighter residual distributions.

Under unconditional conformal construction, a single global quantile threshold is applied to all observations. If conformity score distributions differ substantially across feature regions, global thresholds may lead to:

- Under-coverage in minority or high-risk regions
- Over-coverage in low-risk regions

This phenomenon can exist even in the absence of temporal drift.

Mitigation Strategy:

To address this structural heterogeneity, a natural approach is to introduce stratified conformal constructions, such as class-conditional (Mondrian) conformal prediction and risk-bucketed Mondrian schemes. where quantile thresholds are computed separately within:

- Class labels (to address imbalance)
- Risk-score bins (to stabilize coverage across risk regions).

These stratified constructions do not correct temporal concept drift but instead reduce feature-dependent coverage instability.

B.5 Shift-Aware Conformal Framework

Based on the diagnostic analysis in Section B.4, coverage deviation arises from multiple sources of distributional discrepancy. To improve coverage stability under temporal evaluation, we construct a shift-aware conformal inference framework that integrates three complementary components:

1. Density-ratio reweighting to address covariate shift
2. Rolling calibration to adapt to temporal conditional shift
3. Stratified Mondrian constructions to reduce conditional heterogeneity

These components are designed to target different mechanisms of coverage instability while preserving the distribution-free guarantees of conformal prediction under approximate exchangeability.

B.5.1 Density-Ratio Reweighting for Covariate Shift

To mitigate temporal covariate shift, we incorporate density-ratio reweighting into the conformal calibration process.

Specifically, we estimate the density ratio between test and calibration feature distributions:

$$w(X) = \frac{P_t(X)}{P_s(X)}$$

In practice, these ratios are estimated using a domain classifier trained to distinguish calibration samples from test samples. The predicted probabilities from the classifier are then converted into density-ratio weights.

Given conformity scores s_i and weights w_i , the conformal threshold is obtained using a weighted quantile:

$$q = \inf \left\{ t : \sum_i w_i 1(s_i \leq t) \geq (1-\alpha) \sum_i w_i \right\}$$

This reweighting aligns the effective calibration distribution with the test distribution, thereby reducing coverage distortion induced by covariate shift.

However, density-ratio reweighting only addresses differences in feature distributions and does not account for temporal changes in the risk function.

B.5.2 Rolling Calibration for Temporal Conditional Shift

To mitigate temporal conditional shift, we adopt a rolling calibration scheme.

Rather than estimating conformity score quantiles using a fixed calibration dataset, thresholds are recalibrated using a sliding window consisting of the most recent W weeks of observations.

Formally, for each test week t , calibration scores are computed using data from weeks:

$$t-W, \dots, t-1$$

The conformal quantile is then estimated from this rolling calibration window and applied to week t .

This rolling mechanism allows conformity thresholds to adapt to slowly evolving residual distributions, reducing the mismatch between historical calibration scores and future prediction errors.

It is important to emphasize that rolling calibration mitigates the effects of gradual concept drift but does not eliminate it entirely. Full correction would require model retraining or structural updates to the predictive model.

B.5.3 Risk-Bucketed Mondrian Conformal Prediction

To address conditional heterogeneity across risk regions, we extend the Mondrian conformal construction by introducing risk-based stratification.

Instead of applying a single global quantile threshold, we partition observations into risk buckets based on predicted probabilities:

$$B(X) = \text{bin}(\hat{p}(X))$$

where $\hat{p}(X)$ denotes the predicted default probability.

For each risk bucket b , separate conformal thresholds are estimated:

$$q_b = \text{Quantile}_{1-\alpha} \{s_i : B(X_i) = b\}$$

The prediction set for a new observation is then determined using the threshold corresponding to its risk bucket.

This stratified construction allows different risk regions to have different conformity thresholds, which stabilizes coverage across heterogeneous feature regions.

Importantly, risk-bucketed Mondrian constructions do not correct temporal concept drift. Instead, they mitigate feature-dependent coverage instability that arises when conformity score distributions differ across risk levels.

B.6 Empirical Coverage Improvement

B.6.1 Coverage Stability Over Time

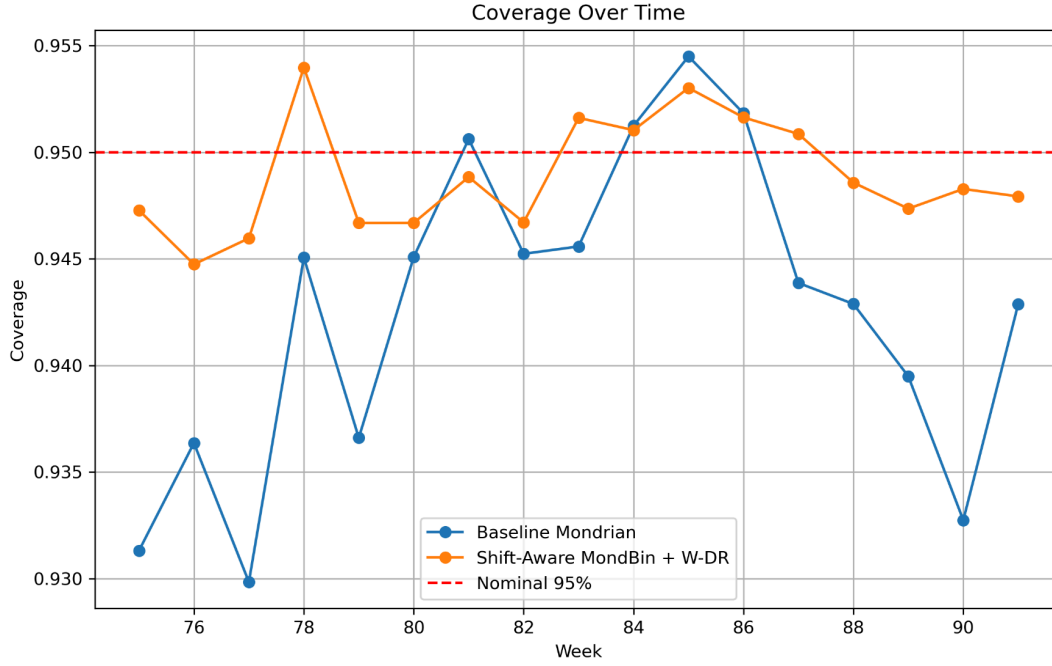


Figure 13 coverage behavior

Figure 13 illustrates the temporal coverage behavior of the baseline Mondrian conformal predictor and the proposed shift-aware framework. The baseline Mondrian construction exhibits noticeable deviations from the nominal 95% level across multiple weeks. In contrast, the shift-aware framework integrating density-ratio reweighting, rolling calibration, and risk-bucketed Mondrian substantially stabilizes coverage over time. The empirical coverage remains consistently closer to the nominal level and exhibits reduced temporal fluctuation.

B.6.2 Overall Coverage Improvement

Method	Coverage	Coverage ($y=1$)	Avg Set Size	Singleton Rate
Baseline Mondrian	0.9430	0.9555	1.445	0.555
Shift-Aware MondBin + W-DR	0.9492	0.9461	1.611	0.365

Table 10. Empirical Coverage Comparison Between Baseline Mondrian and the Proposed Shift-Aware Framework

Table 10 summarizes the aggregate performance across the evaluation period. The baseline Mondrian conformal predictor achieves an overall coverage of **0.943**, remaining below the nominal 0.95 level. After incorporating the shift-aware adjustments, the proposed framework improves overall coverage to **0.949**, bringing empirical coverage substantially closer to the target level while maintaining stable performance across evaluation weeks.

B.6.3 Coverage in the Minority Class

Importantly, the improvement is particularly meaningful for the default class ($Y=1$), which represents the minority but operationally critical segment in credit risk modeling.

While the baseline Mondrian construction achieves slightly higher coverage for the default class (0.955), this comes at the cost of poorer global coverage calibration. The proposed shift-aware framework provides more balanced coverage across classes while improving overall coverage stability.

B.7 Discussion

Although the proposed shift-aware framework substantially improves empirical coverage stability, the overall coverage remains slightly below the nominal 95% level.

Several factors may explain this residual gap.

First, temporal concept shift may still persist beyond the capacity of rolling calibration to fully adapt. While the rolling calibration window allows conformity thresholds to adjust to recent score distributions, it cannot completely correct structural changes in the underlying risk function $P(Y|X)$. Addressing such shifts would require periodic model retraining or structural updates to the predictive model.

Second, density-ratio reweighting only corrects marginal feature distribution discrepancies. If distribution shift involves deeper conditional changes in the relationship between features and default risk, reweighting alone cannot fully restore exchangeability.

Third, real-world credit portfolios often exhibit complex temporal dynamics and heterogeneous risk segments. Even with stratified conformal constructions, perfect finite-sample coverage may be difficult to achieve under persistent non-stationarity.

Despite these limitations, the proposed framework significantly improves empirical coverage stability and brings overall coverage closer to the nominal level while maintaining reasonable prediction set sizes. This demonstrates the practical value of explicitly accounting for distributional discrepancies when deploying conformal prediction in real-world credit risk environments.

